OFFICIAL BIDDING DOCUMENT

Telecommunications Technology Solutions Inc. (TelTech)

Corporate Tender Secretariat · Procurement & Network Engineering Division

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Site Survey, RF Planning & Civil/Power Engineering Guidelines

TowerCo Co-Location Standard, Greenfield Siting & Link Budget Engineering Standards

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RFP-TELTECH-FWA-2026-001	OLIVER TUNGOL	Band n50 (1.5 GHz, 60 MHz TDD)	Official Request for Proposal

1. ENGINEERING PRINCIPLES & SITING ARCHITECTURE

1.1 Siting Philosophy & Mandatory 3-Tier Sourcing Hierarchy

The primary engineering objective of the 5G FWA rollout is to maximize household population coverage within high-density suburban and rural barangay settlement clusters while minimizing civil capital expenditures and permitting lead times. To achieve rapid 1,000-site Phase 1 execution within 12 months, the Contractor shall execute site delivery strictly following the **Mandatory 3-Tier Infrastructure Sourcing Hierarchy**:

1. **Tier 1 (Mandatory Baseline — Existing TowerCo Co-Location):** Field teams must prioritize co-location on existing operational towers (PTCI/MIDC, EdgePoint, FTAP Globe, FTAP Smart, LDIC, Phil-Tower) located within the designated search ring. The Operator leverages an audited master database of **13,746 operational towers**. Co-location reduces Ready-for-Installation (RFI) lead times from 120–180 days down to 15–30 days while eliminating private land titling and environmental permitting complexities.
2. **Tier 2 (Secondary — TowerCo Build-to-Suit / B2S):** If all existing towers within the search ring exceed structural loading thresholds (EPA limits) or fail RF line-of-sight requirements, the vendor shall engage a licensed Independent TowerCo to execute a Build-to-Suit (B2S) under TelTech's Master Lease Agreement (MLA).
3. **Tier 3 (Restricted Exception — Operator Greenfield Own-Build):** Direct operator-constructed greenfield monopoles are strictly limited to remote isolated barangays located > 2.0 km from any operational tower structure (~9.3% of sites nationwide; 103 sites in Phase 1). Proposing an own-build greenfield site requires formal written submission of **Structural Denial / Space Unavailability Letters** from at least two (2) licensed TowerCos and written sign-off from the TelTech Project Steering Committee.

1.2 Search Ring Specification & Demand Centroid

For each nominal site location assigned in Document 07 (Candidate Site List):

- **Urban & Suburban Search Ring:** Defined as a circle with a radius of **500 meters** centered on the official nominal barangay demand centroid.
- **Rural & Remote Search Ring:** Defined as a circle with a radius of **1,000 meters** (expandable up to 1,500 meters upon TelTech RF approval only if prominent terrain ridges or hills provide superior line-of-sight).
- **Target Coverage Guarantee:** Sited candidate coordinates must achieve unobstructed Line-of-Sight (LOS) and target RSRP coverage (≥ -95 dBm) across ≥ 90% of the inhabited settlement zone.

1.3 Mandatory Three-Candidate Rule (Candidates A, B, C)

For every search ring, the Turnkey Contractor / Site Acquisition Vendor must survey, evaluate, and submit **exactly three (3) distinct candidates** in the Technical Site Survey Report (TSSR):

- **Candidate A (BEST — Primary Recommended):** Optimal TowerCo co-location asset with immediate structural capacity at ≥ 30m AGL, available -48V DC busbar or commercial AC transformer headroom, nearest fiber node, and fastest projected RFI (≤ 21 calendar days).
- **Candidate B (BETTER — Secondary Target / Backup 1):** Fully compliant alternative TowerCo structure within the search ring requiring minor solvable modifications (e.g., dedicated AC sub-meter installation, mount bracket extensions, or secondary fiber lead-in route).
- **Candidate C (GOOD — Contingency / Fallback):** Third viable TowerCo asset or TowerCo Build-to-Suit option, ensuring the search ring is not stalled if Candidates A and B encounter unforeseen structural or commercial deadlocks.
- **14-Day Lease / SAA Escalation Protocol:** If the Site Application & Allocation (SAA) or Service Order (SO) negotiation for Candidate A stalls or exceeds **14 calendar days**, the vendor is contractually mandated to immediately activate and lock **Candidate B** to prevent project schedule slippage.

1.4 Co-Location Candidate Evaluation Criteria Matrix (100-Point Scoring)

- **Space & Structural Capacity (20%):** Antenna mounting space at ≥ 30m to 45m AGL, wind load headroom for 3x n50 antennas + 3x 8T8R RRUs (≤ 45 kg each), and ≥ 2.0m × 2.0m compound ground space for outdoor rectifier/battery cabinets.
- **Access Road & Compound Security (15%):** All-weather paved/concrete road access, established TowerCo security fencing/guarding, and 24/7 unobstructed Operations & Maintenance (O&M) ingress/egress.
- **RF Line-of-Sight & Propagation (25%):** Dominant ground elevation AMSL, clear first Fresnel zone into residential settlement, zero close-in clutter shielding within 200m of sector azimuths (0°, 120°, 240°).
- **Backhaul Transport Proximity (15%):** Proximity to Converge/TelTech fiber distribution closure (Class A ≤ 0.5km, Class B ≤ 1.0km), existing TowerCo cable ladder/conduit entry, or unobstructed northern sky view (0° – 80° elevation arc) for Starlink satellite backhaul.
- **Power Availability & Utility Infrastructure (15%):** Commercial utility grid transformer capacity, available -48V DC busbar tap (≥ 63A) or space for dedicated AC service entrance sub-meter, and shared backup genset/battery support.
- **Lease & Commercial Terms (10%):** Fast-track Service Order execution under existing TowerCo Master Lease Agreement (MLA), monthly recurring fee within budget, and guaranteed RFI timeline ≤ 30 calendar days.

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2. DETAILED TECHNICAL SITE SURVEY REPORT (TSSR) REQUIREMENTS

For every nominal site candidate, the Contractor shall complete and submit a comprehensive TSSR conforming to the following structure:

2.1 General Site Identification & Geography

- Site Name, Candidate ID, Official PSGC Code, Region, Province, Municipality, and Barangay.
- Precision WGS84 GPS Coordinates (Latitude, Longitude recorded with >= 6 decimal places via dual-frequency GPS receiver).
- Ground Elevation Above Mean Sea Level (AMSL in meters).
- Access Road Assessment: Road surface type (concrete, asphalt, macadam, dirt trail), bridge weight limits, and 4x4 hauling accessibility during monsoon seasons.

2.2 Tower Structure & Loading Audit (Co-Location Sites)

- Operating TowerCo Entity (PTCI/MIDC, FTAP Globe, EdgePoint, FTAP Smart, LDIC, FTAP BTS).
- Tower Type (Self-Supporting 3-Leg/4-Leg Tower, Monopole, Guyed Mast, Rooftop Tower).
- Tower Height Above Ground Level (AGL in meters).

- Structural Space Audit: Proposed mounting height for 3x Band n50 antennas (recommended centerline: 30m to 45m AGL).
- Structural Loading Verification: Confirmation that existing tower members and foundation can accommodate additional 3x 8T8R RRUs/AAUs (< 45 kg each) and wind load profile of 250 km/h.
- Panoramic Drone Photographic Survey: High-resolution 360° continuous panoramic video and 12-cardinal-point still photos at proposed antenna centerline height to verify optical line-of-sight across populated settlements.

2.3 Power Infrastructure & Utility Audit

- Existing Commercial AC Utility Power: Electric Cooperative / Distribution Utility name (e.g., MERALCO, PELCO, BENECO, CAPELCO, ILPI).
- Service Entrance Capacity: Transformer rating (kVA), service voltage (230V / 400V, 60 Hz), breaker rating, and phase balance.
- TowerCo DC Power Availability: Sourcing from TowerCo DC busbar (-48V DC) vs. installing Operator-dedicated rectifier subrack.
- Sub-metering feasibility and cable tray routing for power cables.
- Grounding System Audit: Measurement of existing earth ground resistance using a 3-point fall-of-potential earth tester. Earth resistance must be <= 5.0 Ohms.

2.4 Backhaul Transmission Path Survey

- Audited Converge Fiber Network Match (subject to verification on the ground): Distance to nearest confirmed node candidate (Class A <= 0.5 km, Class B <= 1.0 km) or line-tap closure (Class C <= 0.5 km).
- Cable Entry Facility: Manhole, handhole, or aerial pole lead-in routing into tower shelter/compound.
- For Remote Satellite Sited Candidates (Starlink Business): Satellite antenna mounting location (unobstructed sky view with 0° to 80° northern elevation arc for Starlink constellation tracking).

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3. RF PROPAGATION & CELL DIMENSIONING GUIDELINES

3.1 Link Budget Parameters (Band n50, 1.5 GHz)

PARAMETER	UNIT	VALUE (DOWNLINK)	VALUE (UPLINK)	TECHNICAL NOTES
Operating Frequency	MHz	1487.0 (Center)	1487.0 (Center)	Band n50 TDD (1457.0 – 1517.0 MHz Channel Range)
Channel Bandwidth	MHz	60.0	60.0	162 Resource Blocks (SCS 30 kHz)
BTS Maximum Conducted Power	dBm / W	55.0 dBm (320 W)	N/A	8 x 40W RRU configuration
BTS Power per Resource Block	dBm	32.9 dBm	N/A	Uniform power distribution
BTS Antenna Gain	dBi	17.5 dBi	17.5 dBi	8-port cross-pol macro antenna
BTS Cable / Jumper Loss	dB	1.0 dB	1.0 dB	Short optical/RF jumper interface
CPE Maximum Transmit Power	dBm / W	N/A	23.0 dBm (200 mW)	Power Class 3 UE (or Class 2 26 dBm)
CPE Antenna Gain	dBi	9.0 dBi	9.0 dBi	Baseline model (8 dBi indoor / 12 dBi outdoor)
CPE Body / Indoor Penetration Loss	dB	8.0 dB	8.0 dB	Concrete hollow block & glass attenuation
Foliage / Clutter Margin	dB	6.0 dB	6.0 dB	Typical rural tropical tree canopy

PARAMETER	UNIT	VALUE (DOWNLINK)	VALUE (UPLINK)	TECHNICAL NOTES
Shadowing Margin (Log-Normal)	dB	7.0 dB	7.0 dB	90% edge cell coverage reliability
Thermal Noise Density	dBm/Hz	-174.0	-174.0	kT @ 290K
Receiver Noise Figure	dB	6.0 dB (CPE)	3.5 dB (gNodeB)	Active RF front-end
Target SINR for Min Service	dB	-1.0 dB	-2.0 dB	Minimum modulation threshold (QPSK 1/3 cell-edge; +10 dB target for residential 64QAM)
Maximum Allowable Path Loss (MAPL)	**dB**	**144.5 dB**	**139.8 dB**	**Uplink is coverage limiting**
Calculated Reliable Cell Radius	**km**	**3.5 km**	**3.2 km**	**Planning model assumption in suburban/rural clutter**

3.2 Nominal Sector Azimuth & Tilt Strategy

- **Sector Count:** 3 sectors per macro site with 120° nominal separation (0°, 120°, 240°). Azimuths must be tailored during TSSR to track actual barangay population concentrations along transport corridors and coastal plains.
- **Electrical Remote Downtilt (eRET):** Antennas must feature AISG 2.0 remote tilt adjustable between 2° and 12° from the central OSS. Mechanical tilt shall only be applied when required to clear close-in obstructions (>15°).

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4. CIVIL WORKS & STRUCTURAL ENGINEERING (CLASS T4 GREENFIELD & TOWERCO BTS SITES)

For sites requiring self-constructed greenfield towers or TowerCo Build-to-Suit (B2S) structures, the Contractor shall adhere to the following civil and structural standards conforming to the **National Structural Code of the Philippines (NSCP 2015 7th Edition, Section 207 Wind Loads)** and **TIA-222-G/H**:

4.1 Philippine Basic Wind Speed Regimes & Wind Zone Mapping

Structural calculations and wind force determinations must strictly comply with NSCP 2015 basic wind speed maps:

- **Wind Zone 1 (Ultimate Wind Speed V = 315 km/h / 87.5 m/s):** Covers the high-typhoon corridor along the eastern seaboard (Batanes, Cagayan, Isabela, Aurora, Quezon/Polillo, Bicol Region, Samar, Leyte, Dinagat Islands, Surigao del Norte/Sur, Davao Oriental).
- **Wind Zone 2 (Ultimate Wind Speed V = 270 km/h / 75.0 m/s):** Covers Central and Western Luzon, National Capital Region (NCR), Central Luzon (Pampanga, Bulacan, Tarlac, Bataan, Zambales, Nueva Ecija), CALABARZON, MIMAROPA, and Western/Central Visayas (Iloilo, Cebu, Bohol, Negros).
- **Wind Zone 3 (Ultimate Wind Speed V = 200 km/h / 55.6 m/s):** Covers low-risk regions including Palawan, Zamboanga Peninsula, and southern inland Mindanao.

4.2 Structural Limit States & Deflection Criteria

- **Survivable Wind Speed ($V_{survival}$ — Ultimate Limit State / LRFD):** Designed for **270 km/h to 315 km/h** 3-second gust wind speed under Ultimate Limit State with full equipment loading (3x n50 antennas, 3x RRUs, satellite dish, and mounting steelwork). Zero plastic collapse or structural overturning.
- **Operational Wind Speed ($V_{operational}$ — Serviceability Limit State / SLS):** Designed for **140 km/h to 160 km/h (40.0 m/s to 44.4 m/s)** continuous operational wind speed (1.0 load factor per TIA-222). Structure maintains rigid RF alignment, ensuring uninterrupted 5G beamforming, MIMO rank-4 transmission, and backhaul link availability (≥ 99.995%).
- **Allowable Twist (θ_{twist} / Torsional Rotation):** ≤ ±0.5° (30 arcmin) at 30m AGL for narrow-beam / microwave dish antennas; ≤ ±1.0° (60 arcmin) at 30m AGL for 5G Band n50 macro sectors.

- **Allowable Sway / Tilt (θ_{sway} / Angular Deflection):** ≤ ±1.0° (60 arcmin) at top antenna mounting centerline (30.0m AGL).
- **Allowable Verticality (Tower Plumbness / Alignment):** ≤ 1 : 500 (0.20% of Tower Height H) throughout the full mast height under dead load and still air. For a 30.0m monopole, maximum horizontal deviation Δ ≤ 60 mm (for 40.0m monopole, Δ ≤ 80 mm). Tolerance between consecutive slip-joints: ≤ 1 : 1000 (0.10%).
- **Horizontal Service Deflection (Δ_{top}):** ≤ H / 100 (300 mm maximum at 30.0m AGL).

4.3 Monopole Tower & Material Specification

- **Tower Structure:** 30-meter or 40-meter polygonal tubular steel monopole constructed in 3 to 4 slip-joint sections with minimum 1.5 × inside diameter overlap.
- **Structural Steel:** ASTM A572 Grade 50 (F_y = 345 MPa / 50 ksi, F_u = 450 MPa).
- **Corrosion Protection:** Hot-dip galvanized per ASTM A123 / ISO 1461 with minimum zinc coating thickness ≥ 85 µm (610 g/m²).
- **Fasteners & Anchors:** Anchor bolts ASTM F1554 Grade 55/105 with double heavy-hex nuts; structural bolts ASTM A325 / ISO 898-1 Class 8.8 hot-dip galvanized.

4.4 Compound Civil Works & Foundation

- **Compound Footprint:** Standard 8.0m × 8.0m (64 m²) gravel-topped compound with geotextile weed barrier.
- **Perimeter Fencing:** 2.4-meter high cyclone wire mesh fence topped with 3 strands of galvanized concertina razor wire.
- **Equipment Pad:** Reinforced concrete equipment plinth (Class A concrete, 28-day compressive strength f'_c ≥ 28 MPa / 4,000 psi, ASTM A615 Grade 60 rebar) for outdoor power/battery cabinets.
- **Foundation:** Reinforced concrete pad and chimney mat foundation or bored piles engineered with Safety Factor ≥ 2.0 against soil bearing failure and ≥ 1.5 against overturning/sliding.

4.5 Lightning Protection & Earthing Grid

- Top-mounted Franklin lightning air terminal extending ≥ 2.0m above uppermost antenna.
- Down-conductor: 70mm² bare stranded copper cable routed externally and clamped every 1.0m to tower shaft.
- Ground ring: Continuous buried bare copper loop around tower base and compound with copper-clad steel ground rods (19mm dia × 3.0m length) exothermically welded (Cadweld) at all junctions, achieving earth ground resistance ≤ 5.0 Ohms.

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5. SINGLE SITE VERIFICATION (SSV) & CLUSTER ACCEPTANCE CRITERIA

Before any site is granted Commercial On-Air Acceptance, the Contractor must complete and submit a signed SSV Report verifying:

1. **Stationary Functional Tests:**

* Sector Verification: Physical azimuth and tilt match TSSR approval within +/- 2°.
* RF Transmit Power: Conducted power on all 8 transmit paths within +/- 0.5 dB of nominal.
* VSWR / Return Loss: Return loss >= 15 dB across all antenna feeder lines.
* Max Throughput Test: Single CPE in line-of-sight achieving >= 300 Mbps DL and >= 50 Mbps UL (aggregate sector peak >= 450 Mbps DL / >= 60 Mbps UL verified under multi-CPE or loaded test).

2. **Short Drive Test & Static Multi-Point Test:**

* 5 stationary measurement points per sector (Near: 500m, Mid: 1.5 km, Far/Edge: 3.0 km).
* RSRP >= -95 dBm across >= 90% of surveyed barangay inhabited zone.
* Ping RTT <= 18 ms target (<= 30 ms maximum contractual limit) to local UPF IP address over fiber backhaul; packet loss <= 0.10% over 1,000 continuous pings.
* Inter-sector handover execution without drop during transition.

(End of Appendix D — Site Survey, RF Planning & Civil Engineering Guidelines)